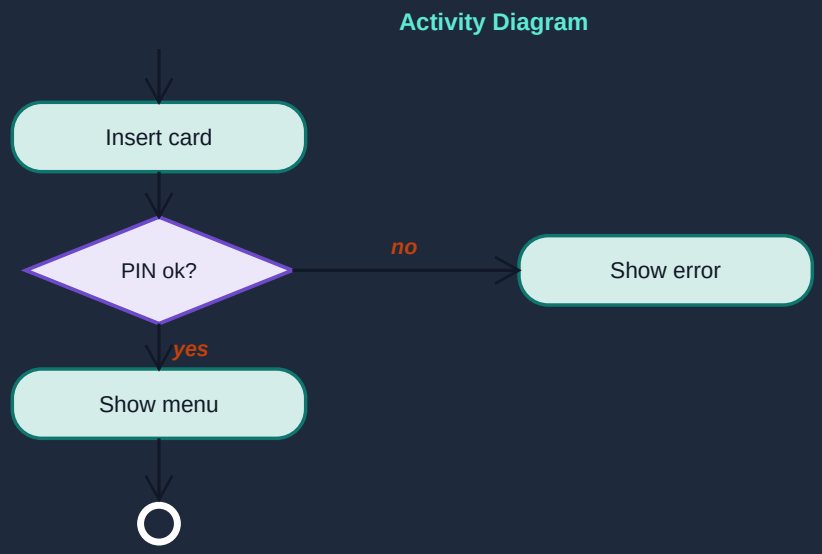
SOFTWARE ENGINEERING | SECTION 5

ERD & Activity Diagram

How a system stores its data, and how its work flows step by step. Example: an ATM.



Prepared by Amira Mahmoud

ERD: Entity Relationship Diagram

- A diagram that shows the DATA a system stores and how the pieces of data are related.
- It is the blueprint of the DATABASE, drawn before creating the tables.
- Entities become tables. Attributes become columns. Relationships become foreign keys.
- It answers: what do we store, and how is it connected?

Think of it as

Excel sheets and the links between them. Each sheet is a table, each column is an attribute.

Entity

A thing we store data about. It becomes a table.

CUSTOMER, ACCOUNT, CARD

Attribute

A property of the entity. It becomes a column.

name, balance, card_number

Relationship

A link between two entities, named with a verb.

Customer owns Account

Class Diagram

Objects with data AND methods. Used to write code.

ERD

Only DATA, no methods. Used to design the database.

ERD Notation: Keys and Cardinality

Crow's foot symbols (placed at the end of the line)

Exactly one

1

Zero or one

0..1 (optional)

One or many

1..*

Zero or many

0..*

How to read the line

One CUSTOMER owns one or many ACCOUNTS. Each ACCOUNT belongs to exactly one CUSTOMER.

Primary Key (PK)

Uniquely identifies each row. Never repeated, never empty. Example: customer_id. Shown bold and underlined.

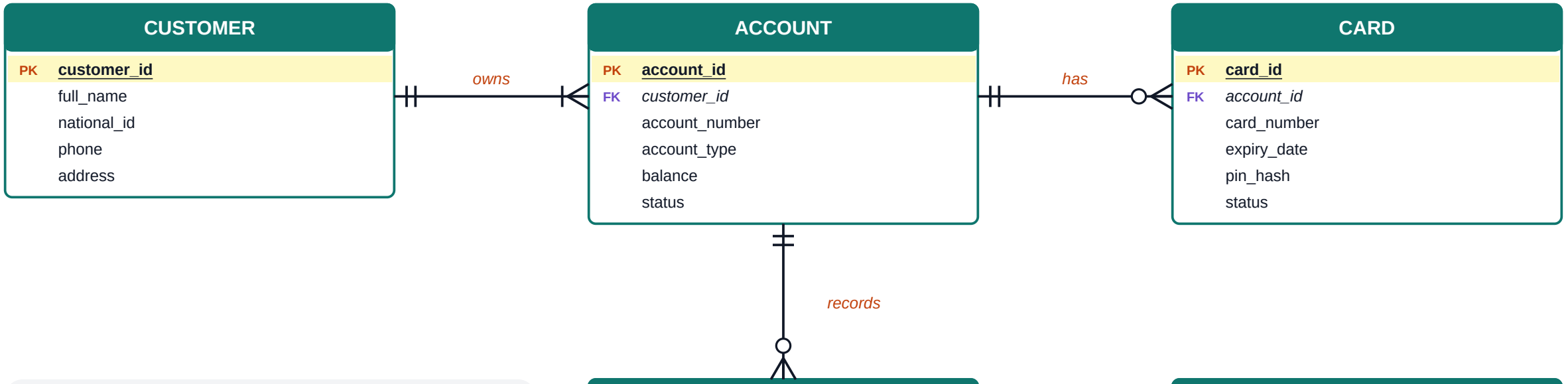
Foreign Key (FK)

A column that holds the PK of another table. It is how two tables are linked. Example: account.customer_id.

Relationship types

One-to-One (1:1)	A Citizen has one ID card.
One-to-Many (1:N)	A Customer has many Accounts.
Many-to-Many (M:N)	Needs a middle table (junction).

ERD Example: ATM System



How to read it

1 Customer owns 1..* Accounts

1 Account has 0..* Cards

1 Account records 0..* Transactions

1 ATM processes 0..* Transactions

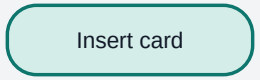
PK = Primary Key **FK = Foreign Key**

Activity Diagram: the Flow of Work

Initial node

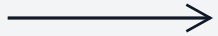
Where the process starts. One only.

Action



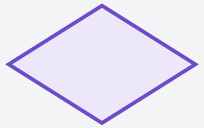
One step of work. Written as a verb.

Flow (arrow)



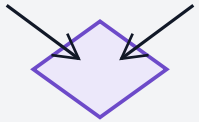
Shows the next step.

Decision



A choice. Each exit has a guard.

Merge



Joins the choices back together.

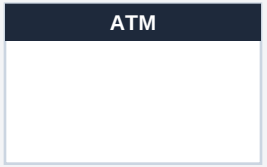
Fork / Join

Fork: steps run at the same time. Join: wait for all.

Final node

Where the process ends.

Swimlane



Who does the step (Customer, ATM, Bank).

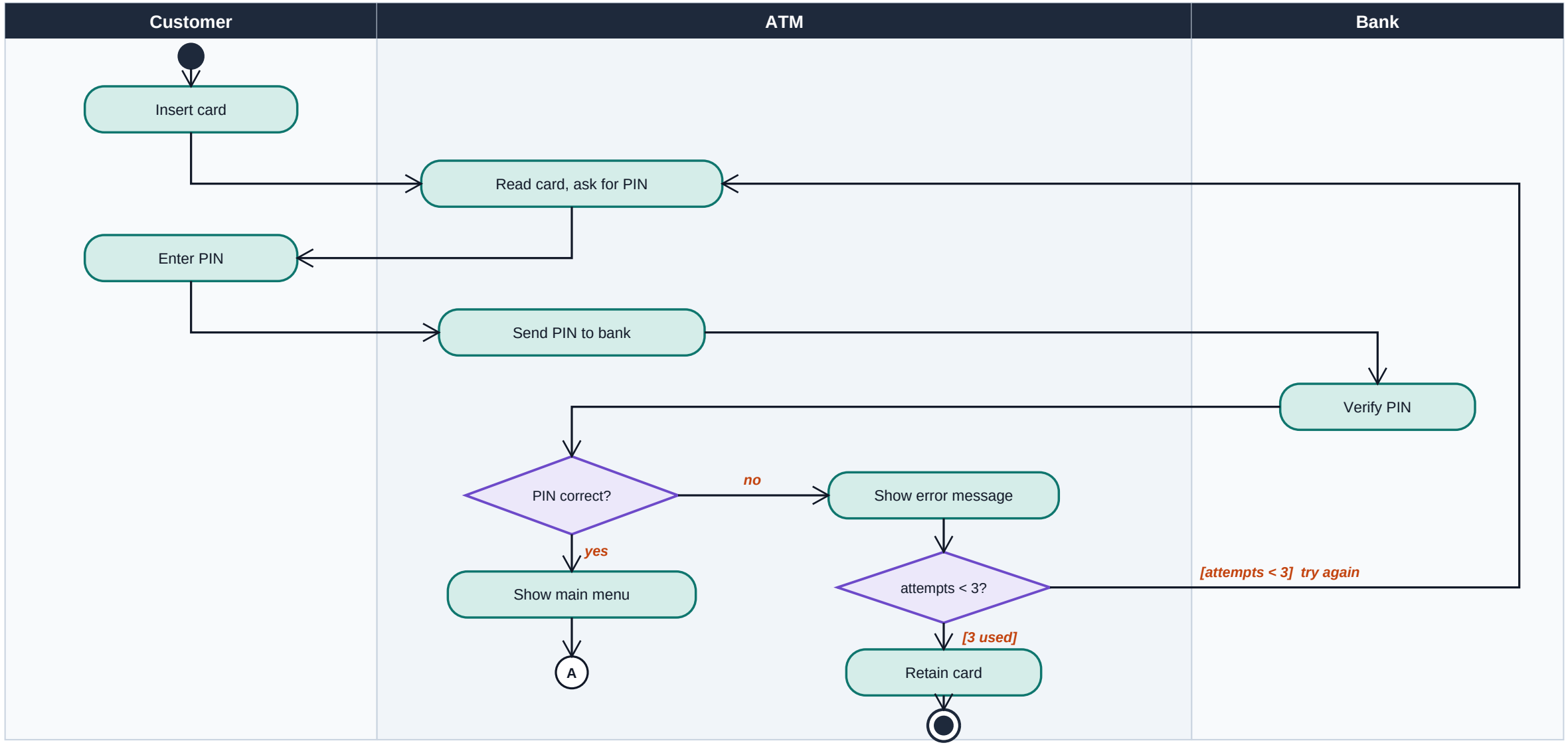
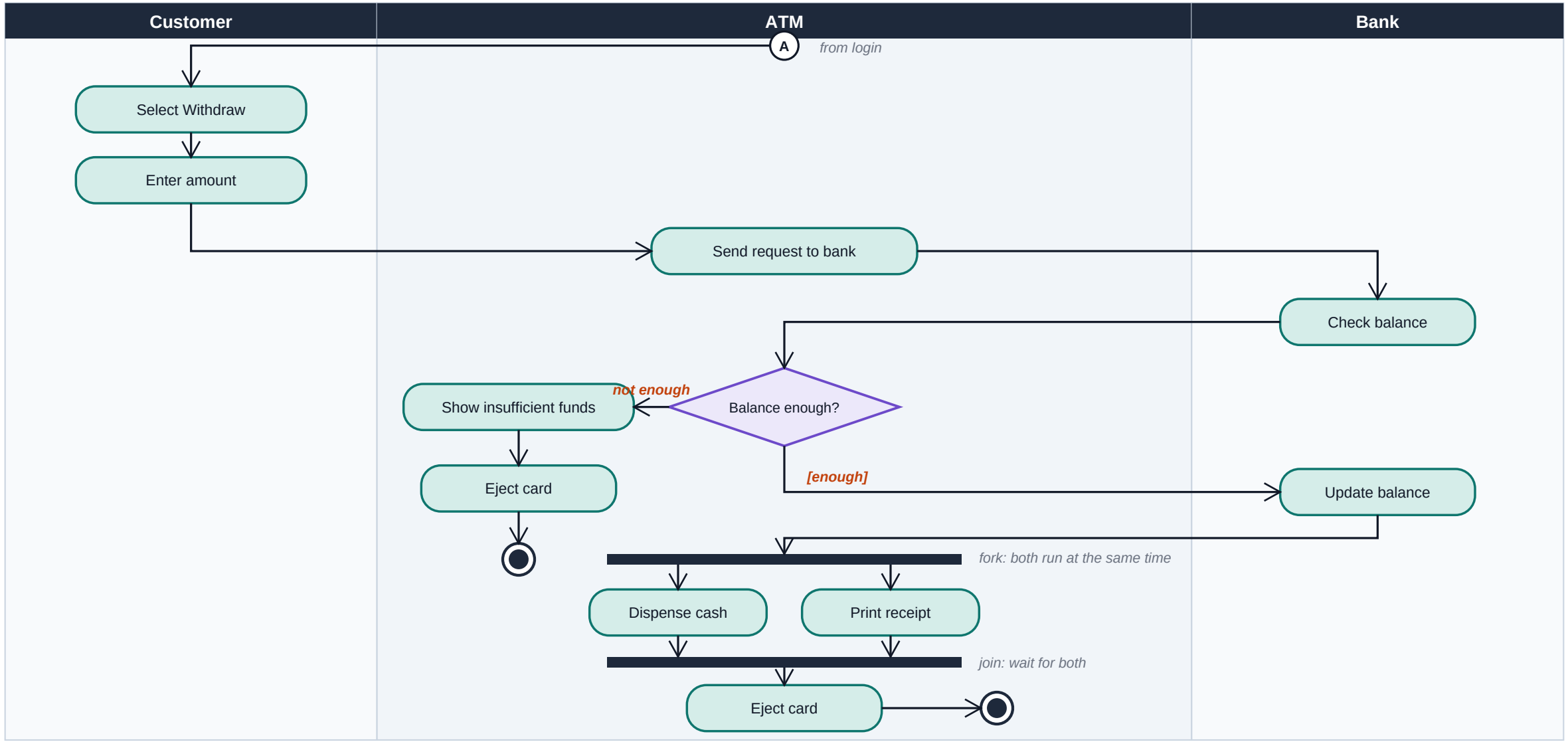
Guard

[PIN is correct]

A condition in brackets: [PIN is correct].

Sequence diagram = who talks to whom. Activity diagram = what happens, in what order, with choices.

Activity Example: ATM Login (PIN check)



Activity Example: Withdraw Cash